

Molecule Space-Time: Predicting Cytochrome P450 Inhibition with a Nonlinear Graph Cellular Automaton

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Abstract

Cytochrome P450 (CYP) inhibition is a major consideration in drug discovery because it can alter drug metabolism and contribute to clinically significant drug–drug interactions. The OpenADMET CYP Inhibition Challenge provides a blinded setting in which to evaluate computational prediction of direct-inhibition pIC₅₀ across four CYP isoforms. Here, we introduce a molecular graph cellular automaton that represents atoms as cells, chemical bonds as local neighbourhoods, and molecular computation as the repeated evolution of a shared, learned transition rule. Unlike molecular models that reduce a structure directly to a fixed representation, our approach retains the complete sequence of atom states and uses its transient and terminal properties to predict CYP inhibition. We call this evolving representation **molecule space-time**: the joint description of molecular structure and its learned progression through computational time. On the sealed scaffold holdout, the cross-fitted target-specific dual-scale Graph-CA ensemble achieved MA-ST-RAE 0.7739 and RMSE 0.8586 pIC₅₀. The corresponding OpenADMET blind evaluation returned MA-ST-RAE 1.0120, macro MAE 1.0861, macro R-squared -0.0766, macro Spearman rho 0.4892, and macro Kendall tau 0.3424. As a secondary objective, we investigated the nonlinear dynamics contained within molecule space-time by extending selected trajectories over thousands of generations and examining convergence, recurrence, periodicity, perturbation sensitivity, strange-attractor candidates, and possible chaotic behaviour. This analysis confirmed two bounded Kuramoto–Sakaguchi trajectories with continually regenerated positive Lyapunov exponents and multidirectional expansion. The framework therefore treats prediction and dynamical exploration as complementary views of the same learned molecular process, offering a route toward CYP inhibition models whose internal evolution can be measured, visualised, and studied as a nonlinear system.

Introduction

The OpenADMET CYP Inhibition Blind Challenge asks participants to predict direct-inhibition pIC₅₀ values for CYP1A2, CYP2C9, CYP2D6, and CYP3A4 from molecular structure, with performance assessed against an unseen test set. Cytochrome P450 enzymes govern the oxidative metabolism of many medicines, and their inhibition can slow drug clearance, increase systemic exposure, alter the metabolism of co-administered compounds, and contribute to clinically important drug–drug interactions. Reliable computational prediction can therefore help medicinal chemists identify liabilities earlier, prioritize experiments, and guide molecular design before more expensive laboratory studies are undertaken.

Our immediate objective in the competition is to produce four useful CYP-specific predictions for every blinded molecule while preserving strict separation between training, model selection, validation, and the hidden challenge set. We approach this as a supervised regression problem in which atom-level chemical descriptors are propagated through the bonded molecular graph, pooled into trajectory fingerprints, and

mapped to pIC50 by a differentiable ridge readout. Several transition rules and temporal scales are combined in an ensemble so that complementary forms of local molecular computation can contribute to the final prediction.

Established approaches to molecular-property prediction include descriptor-based quantitative structure–activity relationship models, random forests and gradient-boosted trees, message-passing graph neural networks, graph transformers, three-dimensional equivariant neural networks, and pretrained molecular foundation models. These architectures provide strong and extensively studied competitive baselines. The present work instead examines a comparatively novel question: whether the aesthetic and dynamical character of information itself can become a useful molecular representation when complex global patterns emerge from repeated local decisions.

We encode each drug as a connected cellular automaton in which atoms are cells, covalent bonds define neighbourhoods, atom descriptors provide the initial information, and a shared transition rule determines how every atom responds to its neighbours at each generation. Bond identity controls the passage of information through single, double, triple, and aromatic connections. This construction retains the defining cellular-automata principle that global behaviour arises from repeated local interactions, while replacing the regular lattice of a classical cellular automaton with the irregular bonded graph of a molecule.

Wolfram's qualitative classification provides a useful vocabulary for the behaviours that cellular automata can produce. Class I systems settle into a homogeneous fixed state; Class II systems form stable or periodic structures; Class III systems generate aperiodic, chaos-like activity; and Class IV systems support persistent localized structures and complex interactions often associated with the boundary between order and disorder. A learned molecular graph cellular automaton does not map automatically onto these lattice-based classes, although the classification motivates measurable analogues including point attractors, oscillations, persistent complex motion, sensitivity to perturbation, and bounded strange attractors.

We regard the method's near-term prospects against highly optimized molecular-learning systems as modest, and its scientific purpose extends beyond leaderboard position. The retained atom-by-channel trajectories allow us to ask how chemical information moves through a ligand, how molecular topology shapes that motion, and whether minute changes to the starting state are repeatedly amplified while the trajectory remains confined to an attracting region. As demonstrated here, particular molecular structures and CYP-conditioned encodings produce bounded trajectories with continually regenerated positive Lyapunov exponents and multidirectional expansion. These strange-attractor dynamics connect molecular structure, local information processing, and emergent behaviour within a single predictive cellular-automata framework.

Materials and Methods

Dataset and Prediction Task

The study used the primary direct-inhibition dataset released for the 2026 OpenADMET CYP Inhibition Blind Challenge. It comprised 4,905 unique compounds represented by a molecule identifier and a SMILES string. Experimental direct-inhibition pIC50 values were provided for four major drug-metabolising cytochrome P450 isoforms: CYP1A2, CYP2C9, CYP2D6, and CYP3A4. Here, pIC50 denotes the negative base-10 logarithm of the half-maximal inhibitory concentration expressed in molar units. Each reported measurement was accompanied by lower and upper uncertainty bounds and an estimated standard deviation from the fitted concentration–response experiment.

The response matrix was incomplete because compounds were not necessarily measured against every CYP isoform. The dataset contained 6,525 observed compound–CYP pairs, distributed as follows:

CYP isoform	Compounds with an observed direct-inhibition pIC50
CYP1A2	1,412
CYP2C9	1,285
CYP2D6	1,493
CYP3A4	2,335
Total	6,525

Each observed compound–CYP pair constituted one supervised regression example. The prediction task was to learn a single CYP-conditioned mapping from molecular structure to direct-inhibition pIC50, rather than fitting an independent model for each isoform. Missing endpoint values were retained as missing and contributed neither targets nor loss terms. The single-concentration, time-dependent-inhibition, and Emax datasets distributed with the challenge were outside the scope of this direct-inhibition study.

To assess generalisation beyond closely related chemistry, compounds were grouped by their standardised Bemis–Murcko scaffold before data partitioning. A fixed 20% subset of scaffold groups was reserved as a sealed holdout, leaving 5,216 observed compound–CYP pairs for model fitting and internal selection and 1,309 pairs for final validation. No scaffold group occurred in both partitions. Hyperparameter selection was conducted within the fitting pool, while the reserved scaffold holdout remained unused until final evaluation.

The challenge test set contained 750 additional compounds for which all direct-inhibition labels were withheld. The trained system was required to generate four finite pIC50 predictions for each compound, giving 3,000 blinded compound–CYP predictions in total. Blinded compounds and their unreleased outcomes were excluded from parameter estimation, hyperparameter selection, early stopping, and dynamical candidate selection. Predictive evaluation followed the challenge formulation: the primary measure was the macro-averaged soft-threshold relative absolute error across the four CYP isoforms, with each isoform weighted equally and predictions falling within the reported experimental uncertainty interval assigned zero error. Conventional regression statistics were retained as complementary measures of predictive agreement.

Molecular Graph Representation

SMILES strings were cleaned with RDKit, reduced to the largest organic fragment, sanitized, and converted to canonical isomeric SMILES. Each standardized molecule was represented as a graph $G = (V, E)$, where every heavy atom $i \in V$ was a cellular-automata cell and every covalent bond defined two directed message-passing edges, $j \rightarrow i$ and $i \rightarrow j$. Molecular geometry was absent from the predictive input.

The baseline atom encoding contained one-hot element identity for H, C, N, O, F, P, S, Cl, Br, I, and other elements; formal charge; aromaticity; sp, sp², sp³, and other hybridization states; degree; total attached hydrogens; ring membership; hydrogen-bond donor and acceptor status; and tetrahedral chirality. Training-only model selection could extend this encoding with five chemically organized feature groups:

Feature group	Atom-level quantities
Periodic	Atomic number, atomic mass, covalent radius, van der Waals radius, outer electrons, period, group, and approximate atomic volume
Valence	Total and implicit valence, heavy-atom degree, radical electrons, and absolute formal charge
Electronic	Pauling electronegativity, approximate polarizability, heteroatom and halogen indicators, conjugated-bond fraction, first ionization energy, electron affinity, and missing-property indicator
Ring geometry	Number of rings containing the atom and indicators for ring sizes 3, 4, 5, 6, 7, and 8 or greater
Local environment	Mean neighbouring electronegativity, electronegativity difference from the neighbourhood, mean neighbouring atomic number, and mean neighbouring formal charge

The selected feature profile was stored with every frozen checkpoint, preserving the exact feature names and order required for inference. Continuous quantities were scaled by fixed chemically meaningful constants during graph construction. Bond vector $\mathbf{e}_{ji}$ contained one-hot single, double, triple, or aromatic identity, conjugation, ring membership, and three stereochemical indicators. The same bond vector was attached to both directed representations of an undirected bond.

The four CYP endpoints were represented by a one-hot context vector $\mathbf{c}$. Consequently, a molecule retained one chemical graph while its cellular-automata trajectory and readout were conditioned on CYP1A2, CYP2C9, CYP2D6, or CYP3A4. Independent molecule–CYP graphs were combined as disconnected components during batched GPU evaluation, so message passing remained confined to atoms belonging to the same molecule.

Nonlinear Graph Cellular Automaton

For atom i , chemical input $\mathbf{x}_i$ was mapped to an initial state with $\mathbf{H}$ dynamical channels:

$$\mathbf{h}_i^{(0)} = \tanh!(\mathbf{s}_0 \mathbf{W}_{\text{init}} \mathbf{x}_i),$$

where $\mathbf{s}_0$ controlled the initialization scale. A shared local rule was then applied recurrently for $\mathbf{T}$ generations. The parameters of this rule were tied across atoms and generations, making the construction a graph recurrent neural network with cellular-automata locality.

At generation t , the message from neighbour j to atom i was

$$g_{ji} = \sigma! \left(\frac{\mathbf{W}_g \mathbf{e}_{ji}}{\theta_b} \right), \quad \mathbf{m}_{ji}^{(t)} = g_{ji} \odot \mathbf{W}_n \mathbf{h}_j^{(t)} + \mathbf{W}_e \mathbf{e}_{ji},$$

where $\mathbf{W}_n$ transformed the neighbouring state, $\mathbf{W}_e$ transformed the bond description, $\mathbf{W}_g$ generated a channel-wise bond gate, θ_b was the bond temperature, and $\odot$ denoted elementwise multiplication. Incoming messages and neighbouring states were degree-normalized:

$$\mathbf{a}_i^{(t)} = \frac{1}{|N(i)|} \sum_{j \in N(i)} \mathbf{m}_{ji}^{(t)}, \quad \bar{\mathbf{h}}_i^{(t)} = \frac{1}{|N(i)|} \sum_{j \in N(i)} \mathbf{h}_j^{(t)}.$$

Chemical identity and CYP context entered every generation through a common reaction drive:

$$\mathbf{r}_i^{(t)} = \tanh! \left(\mathbf{W}_s \mathbf{h}_i^{(t)} + \mathbf{a}_i^{(t)} + \mathbf{W}_x \mathbf{x}_i + \mathbf{W}_c \mathbf{c} + \mathbf{b} \right).$$

The submitted ensemble used five transition rules sharing this bonded message and reaction calculation.

Gated residual. A learned channel-wise gate controlled the proportion of the proposed reaction accepted at each atom:

$$\alpha_i^{(t)} = \sigma! \left(W_\alpha [h_i^{(t)}, a_i^{(t)}, x_i, c] \right), \quad h_i^{(t+1)} = (1 - s\alpha_i^{(t)}) \odot h_i^{(t)} + s\alpha_i^{(t)} \odot r_i^{(t)},$$

where $s\alpha$ was capped at one and s was the update scale.

Inertial reaction–diffusion. A velocity state introduced momentum, while neighbour exchange and restoring forces supplied graph diffusion and damping:

$$f_i^{(t)} = r_i^{(t)} + D \odot (\bar{h}_i^{(t)} - h_i^{(t)}) - R \odot h_i^{(t)}, \\ v_i^{(t+1)} = \eta\gamma \odot v_i^{(t)} + \delta \odot f_i^{(t)}, \quad h_i^{(t+1)} = \tanh! \left(h_i^{(t)} + \delta \odot v_i^{(t+1)} \right).$$

The channel-wise damping γ , step size δ , diffusion D , restoring strength R , and inertial multiplier η were constrained to stable ranges by sigmoid or softplus transformations.

FitzHugh–Nagumo. The state was divided into excitation u and recovery v channels. Their update combined the cubic excitable-system dynamics with learned chemical drive and graph diffusion:

$$\Delta u_i = u_i - \frac{u_i^3}{3} - v_i + \kappa_s \tanh(W_u r_i) + D_u(\bar{u}_i - u_i), \\ \Delta v_i = \epsilon [u_i + q - v_i + 0.1 \tanh(W_v r_i)] + D_v(\bar{v}_i - v_i),$$

The two channel groups were then advanced together:

$$h_i^{(t+1)} = \tanh! ([u_i + s\Delta u_i, v_i + s\Delta v_i]).$$

Kuramoto–Sakaguchi. Each channel was treated as a wrapped phase $\phi_i = \pi h_i$. Bond-gated phase coupling and a chemically conditioned natural frequency gave

$$q_i^{(t)} = \frac{1}{|N(i)|} \sum_{j \in N(i)} g_{ji} \odot \sin! \left(\phi_j^{(t)} - \phi_i^{(t)} - \psi \right), \\ \phi_i^{(t+1)} = \phi_i^{(t)} + s \left[\omega_i^{(t)} + K q_i^{(t)} \right], \quad \omega_i^{(t)} = A \tanh(W_\omega r_i^{(t)}).$$

The updated phase was wrapped and divided by π to return it to $[-1, 1]$. The phase lag ψ , coupling K , and frequency scale A were selected during training-only hyperparameter search.

Delayed memory. A rule-specific delay d selected a preceding state from the retained trajectory. The new drive combined the current reaction, a learned transformation of the current and delayed states, and explicit delayed-state feedback:

$$\tilde{r}_i^{(t)} = \tanh! \left(W_d [r_i^{(t)}, h_i^{(t-d)}] \right), \\ h_i^{(t+1)} = \tanh! \left((1 - \zeta s) h_i^{(t)} + s \left[(1 - \mu) r_i^{(t)} + \mu \tilde{r}_i^{(t)} + \kappa_d (h_i^{(t-d)} - h_i^{(t)}) \right] \right).$$

The delay, memory mixture μ , delayed feedback κ_d , and damping ζ were determined from the rule-specific search space.

Each trajectory was pooled into a molecular fingerprint containing the final atom-state mean and variance, the time-averaged atom state, the temporal variance of the molecular mean state, and mean state-change energy. The multiscale variant appended molecular mean states at 12.5%, 25%, 50%, 75%, and 100% of the trajectory. CYP-specific readout features were formed by combining the endpoint one-hot vector with endpoint-gated copies of the dynamical fingerprint.

Machine-Learning Training and Validation

The submitted Cross-Fitted Target-Specific Dual-Scale Graph Cellular Automata Ensemble, abbreviated CFT-DS-GCAE, was assembled from frozen experts trained under the scaffold partitions described above. Each nonlinear Graph-CA expert was optimized by backpropagation through time. Adam updated the initialization, message, reaction, and transition-rule parameters with a cosine learning-rate schedule, gradient clipping, and an L2 penalty on cellular-automata weights. Generation count, dynamical-channel count, atom-feature profile, learning rate, ridge penalty, update scale, support fraction, batch size, bond temperature, initialization scale and noise, pooling design, and rule-specific dynamical constants were selected using labelled development data alone.

Differentiable ridge readout

Training batches were divided by molecule into support and query subsets. The Graph-CA generated fingerprint matrix $\mathbf{F}_s$ for the support molecules, which was standardized column-wise to $\mathbf{Z}_s$. With centered targets $\mathbf{y}_s - \bar{\mathbf{y}}_s$, the ridge coefficients were obtained by the closed-form differentiable solve

$$\beta = (\mathbf{Z}_s^T \mathbf{Z}_s + \lambda \mathbf{I})^{-1} \mathbf{Z}_s^T (\mathbf{y}_s - \bar{\mathbf{y}}_s).$$

For query fingerprint $\mathbf{f}_q$, prediction was

$$\hat{\mathbf{y}}_q = \bar{\mathbf{y}}_s + \left(\frac{\mathbf{f}_q - \bar{\mathbf{F}}_s}{\mathbf{s}_F} \right)^T \beta,$$

where $\bar{\mathbf{F}}_s$ and $\mathbf{s}_F$ were the support feature mean and scale. The intercept was excluded from the ridge penalty. The linear solve remained connected to the Graph-CA computation graph, allowing query loss gradients to pass through β and into every recurrent generation. A Hermitian pseudoinverse implemented the same ridge objective when highly correlated trajectory statistics made the normal equations numerically singular.

This support–query construction trained the nonlinear cellular automaton to produce fingerprints that generalized beyond the observations used to solve the current ridge layer. At the end of training, a final ridge state was fitted from all permitted fitting observations and stored with the selected Graph-CA checkpoint. Early stopping and checkpoint promotion used MA-ST-RAE on the relevant scaffold-held-out development fold. RMSE was recorded as a secondary optimization diagnostic.

Dual-scale expert ensemble

Five transition-rule families entered the submission: gated residual, delayed memory, inertial reaction–diffusion, Kuramoto–Sakaguchi, and FitzHugh–Nagumo. The original expert family contributed one frozen seed from each rule. The multiscale family averaged seeds 1701, 2909, and 4211 within each rule before combining rules. This design yielded ten expert signals per molecule–CYP pair: five original predictions and five seed-averaged multiscale predictions, derived from 20 frozen checkpoint evaluations.

The first submitted method, DS-GCAE, combined these signals through a fixed global blend. The original family contributed 42.5% of the final prediction and assigned equal weight to its five rules. The multiscale family contributed 57.5% and used rule weights of 0.1036, 0.1999, 0.1966, 0.2447, and 0.2552 for gated residual, delayed memory, inertial reaction–diffusion, Kuramoto–Sakaguchi, and FitzHugh–Nagumo, respectively. These weights were shared across all four CYP endpoints. Thus, DS-GCAE and CFT-DS-GCAE used the same underlying bonded-graph cellular-automata experts and differed in how their frozen predictions were combined.

Cross-fitted target-specific stacking

A separate standardized ridge stack was fitted for each CYP endpoint using the ten expert signals. Ridge penalties **0.01, 0.1, 1, 10, 100, and 1000** were compared inside nested scaffold-grouped folds using endpoint ST-RAE. Each outer-fold prediction was generated by a stack whose penalty and parameters had been selected without that fold. An optional affine calibration

$$\hat{y}_{\text{cal}} = a\hat{y} + b$$

was assessed with calibration penalties of 0, 1, 10, 100, 1000, and an identity option. Calibration training also used out-of-fold predictions. The selected final penalties were 1000 for CYP1A2 and CYP2D6 and 100 for CYP2C9 and CYP3A4. Identity calibration, $a = 1$ and $b = 0$, was selected for all four endpoints.

Final validation and blinded inference

Model development used five scaffold-grouped folds within the fitting pool. The sealed holdout defined above was opened once after expert and stack selection. Evaluation used the primary and complementary metrics specified in the Dataset and Prediction Task subsection, with final uncertainty estimated from 1,000 bootstrap resamples.

After model freezing, the 20 expert checkpoints generated ten aligned signals for each blinded molecule–CYP pair. Four target-specific stacks produced the continuous pIC50 values, and the selected identity calibrations left them unchanged.

Long-Horizon Dynamical Analysis

Targeted discovery of candidate attractor regimes

Dynamical analysis was performed after predictive training, using frozen model parameters and fixed molecular graphs. Candidate selection therefore changed neither the regression model nor its validation score. The objective was to concentrate long-horizon computation on trajectories showing sustained motion, recurrent geometry, broad frequency content, and sensitivity to small perturbations.

An initial screen was calculated from the complete atom-by-channel validation trajectories. For trajectory $\mathbf{H}_t$, the molecular mean state was calculated at each generation, and late-time motion was measured from the mean Euclidean step length over the second half of the observed sequence. Recurrence was assessed over lags from 2 to 64 generations by comparing the mean lagged state distance with the distance expected from the accumulated mean step length. The smallest normalized lagged distance defined the recurrence ratio. Spectral entropy was calculated from the non-zero-frequency Fourier power of each state channel and averaged across channels. A screening score combined these quantities:

$$S = \frac{v_{\text{late}} (1 + H_{\text{spectral}})}{R_{\text{recurrence}}},$$

where $\mathbf{v}_{\text{late}}$ is late-time motion, H_{spectral} is normalized spectral entropy, and $R_{\text{recurrence}}$ is the recurrence ratio. This score acted as a computational targeting device rather than a definition of chaos. Candidates were drawn from gated residual, delayed memory, inertial reaction–diffusion, Kuramoto–Sakaguchi, and FitzHugh–Nagumo transition-rule families so that the long-horizon analysis included contracting, oscillatory, recurrent, and expanding regimes.

Twenty complete molecular states were propagated through 5,000 frozen cellular-automata generations. Every atom and every dynamical channel was retained at every generation. The first 1,000 generations were treated as burn-in for late-time diagnostics. Ten representative cases were subjected to detailed phase-space analysis, including principal-component projections, recurrence plots, frequency spectra, correlation-dimension estimates, and nearest-neighbour divergence curves. Principal-component coordinates were used exclusively for visualization; all perturbation and Lyapunov calculations operated in the full atom-by-channel state space.

Direct perturbation screen

Each detailed case was paired with eight independently oriented full-state perturbations of Euclidean magnitude 10^{-5} . Reference and companion states were advanced under the same frozen transition rule, molecular graph, bond features, atom features, and CYP context. Their separation was recorded across the trajectory. Replicated early separation across perturbation directions was used to prioritize candidates for renormalized analysis. Trajectories 7 and 8, corresponding to OCNT-0494110 conditioned on CYP2C9 and OCNT-2328784 conditioned on CYP1A2, showed the clearest replicated expanding response and were advanced to the confirmatory tests below.

Circular state-space distance

The Kuramoto–Sakaguchi state is phase-like and wrapped to the interval $[-1, 1]$. Differences were consequently measured on the circular state space. For reference state $\mathbf{h}$ and companion state $\mathbf{h}'$, the elementwise circular difference was

$$\Delta(\mathbf{h}', \mathbf{h}) = \frac{1}{\pi} \text{atan2}[\sin(\pi(\mathbf{h}' - \mathbf{h})), \cos(\pi(\mathbf{h}' - \mathbf{h}))],$$

and full-state separation was $d = \|\Delta(\mathbf{h}', \mathbf{h})\|_2$. This prevented an apparent jump across the phase boundary from being interpreted as physical divergence.

Confirmatory protocols

The numerical settings for the confirmatory and population experiments are summarized below. A repeat denotes an independently oriented initial perturbation or orthogonal perturbation basis, as appropriate to the calculation.

Analysis	Burn-in generations	Measured generations	Renormalization or sampling interval	Perturbation scale	Repeats
Largest Lyapunov exponent	1,000	4,000	10	10^{-4} , 10^{-5} , and 10^{-6}	8 per scale and molecule
Float64 Lyapunov spectrum	1,000	4,000	5, 10, and 20	10^{-7}	2 per interval and molecule
Attraction basin	1,000	6,000	Sampled every 10	Displacement radii 0.1, 0.5, 1.0, and 2.0	8 per radius and molecule
Population and structural interventions	1,000	2,000	10	10^{-7}	2 per system

Renormalized largest Lyapunov exponent

Persistent local instability was tested with a Benettin-style repeated-renormalization calculation. A companion state was placed at distance ϵ from the post-burn-in reference state, and both states were advanced for each interval τ . The circular separation d_k was measured, its logarithmic expansion was recorded, and the companion was returned to distance ϵ along the observed separation direction. The largest Lyapunov exponent was estimated as

$$\lambda_1 = \frac{1}{K\tau} \sum_{k=1}^K \log \left(\frac{d_k}{\epsilon} \right).$$

The complete design produced 48 estimates. Repeated renormalization tested whether divergence was continually regenerated after local separations had been returned to the same small scale.

Float64 Lyapunov spectrum

The leading Lyapunov spectrum was calculated in double precision using eight simultaneous orthogonal perturbation vectors. After each propagation block, the full-state circular difference vectors were assembled into a matrix and subjected to reduced QR decomposition. The logarithms of the absolute diagonal elements of the resulting upper-triangular matrix supplied the local expansion rates, while the orthonormal basis supplied the renormalized companion directions. Stability across the intervals and repeats listed above was used to distinguish persistent multidirectional expansion from numerical precision effects or a single unstable direction.

Boundedness and attraction-basin test

Attraction towards a common invariant set was tested by initiating float64 trajectories at the full-state displacement radii listed above. The complete design contained 64 displaced trajectories.

Boundedness was monitored directly from the maximum absolute state. Convergence at the distributional level was evaluated after circular embedding of every state as concatenated sine and cosine coordinates. Distances between early and late trajectory distributions and the reference invariant distribution were estimated over 32 random one-dimensional projections, providing a sliced distribution distance suitable for the high-dimensional state space. A late-to-early distance ratio below one indicated movement towards the reference distribution. Nearest-reference-cloud distance was retained as a complementary finite-sampling diagnostic.

Population-level structure–dynamics analysis

The candidate analysis was followed by a broader test of whether molecular structure was associated with the strength of the learned instability. A scaffold-held-out population of 256 molecule–CYP cases was sampled evenly across CYP1A2, CYP2C9, CYP2D6, and CYP3A4, with 64 cases per endpoint. Sampling was stratified across quartiles of the initial dynamical screening score. The two established leading candidates were added explicitly, producing 258 evaluated cases under the population protocol above.

Molecular descriptors comprised molecular weight, calculated logP, topological polar surface area, hydrogen-bond donors and acceptors, rotatable bonds, ring composition, aromatic and heteroatom fractions, formal charge, fraction sp³, and bond-type counts. Graph descriptors comprised atom and bond counts, cyclomatic number, density, degree moments, adjacency spectral radius, algebraic connectivity, largest Laplacian eigenvalue, graph diameter, and mean shortest-path length. Reproducible ETKDG conformers were generated with seed 260830 and MMFF relaxation to calculate radius of gyration, asphericity, eccentricity, inertial shape factor, and sphericity. Cartesian coordinates were absent from the Graph-CA input, so three-dimensional descriptors were interpreted as structural correlates rather than direct dynamical inputs.

Univariate associations with the largest Lyapunov exponent were measured using Spearman correlation and Benjamini–Hochberg correction within each analysis scope. Confidence intervals were calculated from 2,000 bootstrap resamples of complete Bemis–Murcko scaffold clusters. Multivariate reproducibility was assessed with five-fold scaffold-grouped cross-validation using Elastic Net and Extra Trees regression. Held-out permutation importance quantified the contribution of each descriptor while preserving scaffold separation between training and evaluation folds.

Frozen-model structural interventions

Causal computational tests were performed on trajectories 7 and 8 while retaining all learned weights, the transition rule, the CYP context, and all parameters of the Lyapunov calculation. Each undirected molecular bond was represented by two directed message-passing edges. Bond deletion removed both directions of a selected connection. Bond-identity interventions replaced its single, double, triple, or aromatic edge encoding with each alternative identity. Ring-opening interventions removed a selected ring connection and suppressed the ring-membership atom encoding. Atom-feature interventions ablated one chemically interpretable group at a time: elemental identity, charge and aromaticity, hybridization, local valence, donor–acceptor state, chirality, or neighbouring-atom chemistry.

The intervention campaign comprised 187 modified and baseline systems evaluated with the population protocol. Effect size was defined as the change in the mean largest Lyapunov exponent relative to the corresponding unmodified molecule–CYP baseline. Positive values indicated faster exponential divergence, while negative values indicated slower divergence. The complete workflow, retained numerical tables, and publication figures are available in the [structure–dynamics campaign archive](#).

Results and Discussion

CYP pIC₅₀ Predictions

Sealed internal validation

The cross-fitted target-specific dual-scale Graph Cellular Automata Ensemble (CFT-DS-GCAE) was evaluated once on the sealed scaffold holdout containing 1,309 molecule–CYP observations. Its point MA-ST-RAE was 0.7739. Across 1,000 bootstrap resamples, mean MA-ST-RAE was 0.7749 with a 95% interval from 0.7356 to

0.8193. RMSE was 0.8586 pIC50. The complementary bootstrap macro metrics were MAE 0.6323 pIC50, R-squared 0.2754, Spearman rho 0.5184, and Kendall tau 0.3699.

Metric	CFT-DS-GCAE	Previous DS-GCAE
Point MA-ST-RAE	0.7739	0.7842
Bootstrap mean MA-ST-RAE	0.7749	0.7850
95% bootstrap interval	0.7356–0.8193	0.7464–0.8274
RMSE, pIC50	0.8586	0.8678
Bootstrap macro MAE, pIC50	0.6323	0.6387
Bootstrap macro R-squared	0.2754	0.2706
Bootstrap macro Spearman rho	0.5184	0.5098
Bootstrap macro Kendall tau	0.3699	0.3626

The four target-specific stacks produced point ST-RAE values of 0.8172 for CYP1A2, 0.7611 for CYP2C9, 0.9409 for CYP2D6, and 0.5764 for CYP3A4. The largest residual difficulty on the sealed holdout therefore occurred for CYP2D6, while CYP3A4 gave the strongest endpoint result.

OpenADMET blind challenge evaluation

The challenge organisers calculated the official metrics after submission against labels that remained unavailable during model development. The first submission used the dual-scale Graph-CA ensemble (DS-GCAE) and was recorded at rank 80 of 89. The second used CFT-DS-GCAE and was recorded at rank 82 of 90. Because leaderboard membership changed between snapshots, rank differences do not provide a controlled paired comparison; the metric values supply the direct comparison between the two submitted prediction files.

Official blind metric	DS-GCAE, submission 1	CFT-DS-GCAE, submission 2	Change in submission 2
MA-ST-RAE	1.0132	1.0120	-0.0012
Macro MAE	1.0893	1.0861	-0.0032
Macro R-squared	-0.0827	-0.0766	+0.0061
Macro Spearman rho	0.4751	0.4892	+0.0141
Macro Kendall tau	0.3323	0.3424	+0.0101

Target-specific cross-fitted stacking improved every reported blind metric, with the clearest gains in rank correlation. The blind MA-ST-RAE remained substantially higher than the sealed internal value, and the negative blind macro R-squared indicates that predictive calibration and generalisation across the hidden chemical distribution remain important limitations of the present model. These leaderboard results represent externally calculated challenge outcomes rather than metrics reconstructed from locally available labels.

Nonlinear Dynamics in Molecular Space-Time

Four Dynamical Regimes in Graph Cellular Automata

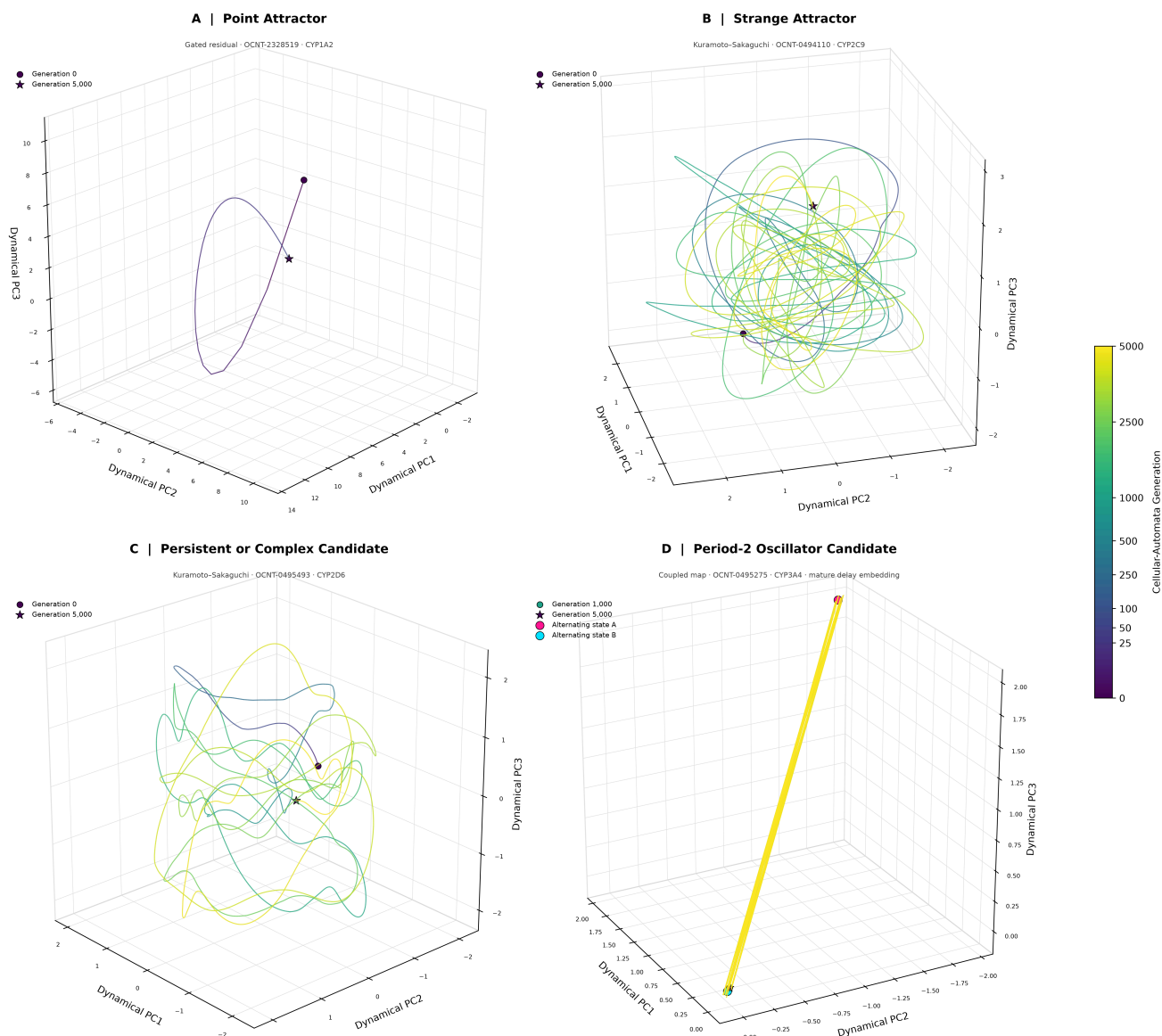


Figure 1. Four long-horizon Graph-CA dynamical behaviours. **A**, the gated-residual trajectory of `OCNT-2328519` conditioned on `CYP1A2` contracts towards a point attractor. **B**, the confirmed Kuramoto–Sakaguchi strange attractor for `OCNT-0494110` conditioned on `CYP2C9` continues to explore a bounded recurrent region. **C**, the Kuramoto–Sakaguchi trajectory of `OCNT-0495493` conditioned on `CYP2D6` forms a persistent or complex candidate with late motion 0.148, spectral entropy 0.605, a dominant timescale of approximately 572 generations, and positive finite-time divergence. This candidate has not undergone the definitive Lyapunov-spectrum and basin-replication tests applied to panel B. Kuramoto phase channels were circularly embedded before projection. **D**, the coupled-map trajectory of `OCNT-0495275` conditioned on `CYP3A4` forms a period-two oscillator candidate with spectral concentration 0.802. The mature oscillator is shown from generation 1,000 through generation 5,000 using a three-coordinate delay embedding of its leading full-state component; magenta and cyan markers identify its alternating states. Panels A–C contain generations 0 through 5,000. Colour records computational time using the colourblind-safe Viridis scale. These coordinates visualize learned atom-by-channel state-space dynamics rather than Cartesian molecular motion. Cyberpunk animations show the [point attractor](#), [strange attractor](#), [persistent or complex candidate](#), and [period-two oscillator candidate](#).

Independent Evidence for Graph-CA Strange Attractors

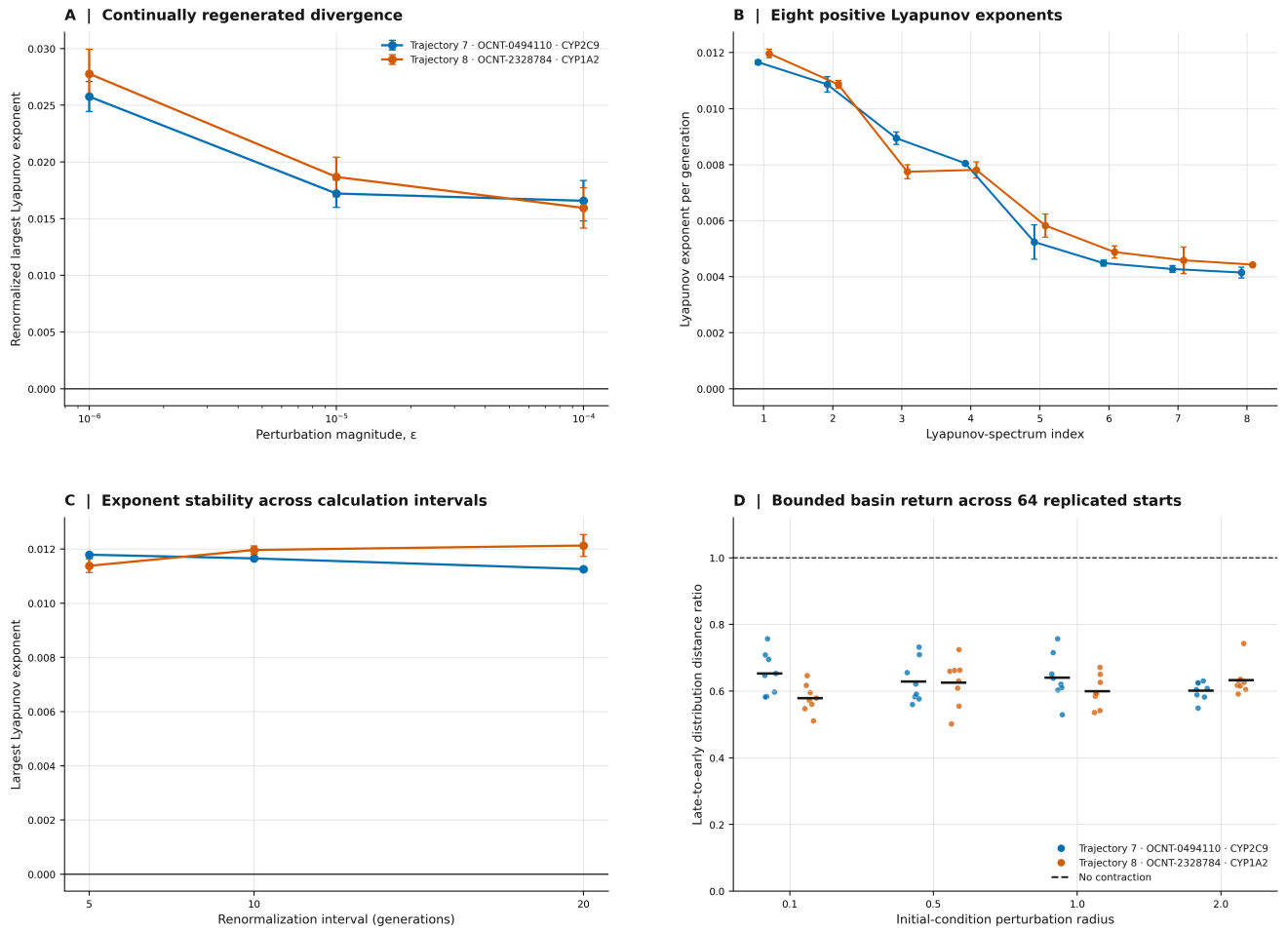


Figure 2. Independent evidence for the Graph-CA strange-attractor classification. **A**, repeated perturbation and renormalization continually regenerated positive largest Lyapunov exponents at perturbation magnitudes from 10^{-6} to 10^{-4} ; all 48 replicated estimates were positive. **B**, the first eight Lyapunov exponents were positive for both definitively tested trajectories, indicating expansion along several independent state-space directions. **C**, the largest exponent remained positive and stable when the renormalization interval was changed from 5 to 10 and 20 generations. **D**, all 64 replicated initial-condition perturbations remained bounded, while late-to-early distribution-distance ratios below one showed approach towards the attracting distribution across four perturbation radii. Error bars in panels A–C show standard deviations across repeated calculations; horizontal black lines mark zero divergence, and the dashed line in panel D marks equal late and early distance. Bounded basin behaviour together with continually regenerated sensitive dependence and a positive Lyapunov spectrum satisfies the operational criteria used here for a hyperchaotic strange attractor. A vector version is available as [PDF](#).

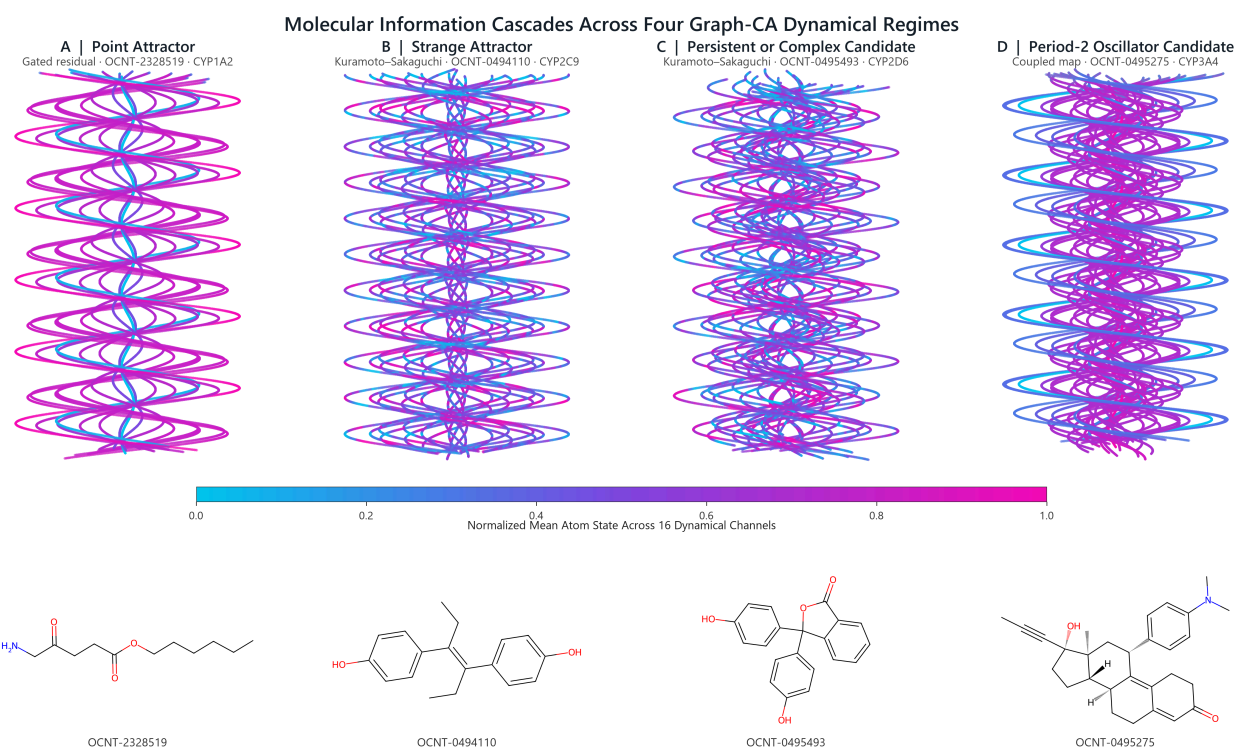


Figure 3. Molecular information cascades generated by the four Graph-CA trajectories in Figure 1. Each panel shows the complete retained cascade at generation 5,000, with the corresponding RDKit 2D molecular structure shown directly beneath it. At every generation, the molecule was rotated slightly and displaced downwards while all earlier atom positions were retained, producing the continuous atom ribbons used in the accompanying molecular space-time videos. Cyan-to-magenta colour records the contemporaneous mean atom state across the 16 dynamical channels after robust normalization within each trajectory. The cascade therefore depicts learned chemical-information evolution across the bonded molecular graph rather than molecular translation through physical space. **A**, gated-residual point attractor. **B**, confirmed Kuramoto–Sakaguchi strange attractor. **C**, Kuramoto–Sakaguchi persistent or complex candidate. **D**, coupled-map period-two oscillator candidate. Bonds are omitted from the cascades to reveal the accumulated atom-state structure. A high-resolution PDF version is available [here](#).

Population-level dynamical screen

The final production model for each transition rule retained dynamical summaries for 1,309 validation trajectories, giving 13,090 short-horizon screens. Point-attractor candidates satisfied late motion below 10^{-4} and a final step below 10^{-5} . Oscillator candidates combined recurrence ratio below 0.25 with spectral concentration above 0.5. Remaining trajectories were assigned to a persistent or complex screening class. These three columns are mutually exclusive. Strange-attractor confirmation was evaluated separately using the 5,000-generation perturbation, renormalized divergence, Lyapunov-spectrum, and basin-replication protocol.

Transition rule	Screened	Point-attractor candidates	Oscillator candidates	Persistent or complex	Confirmed strange attractors
Gated residual	1,309	0	0	1,309	0
Delayed memory	1,309	0	0	1,309	0
Inertial reaction–diffusion	1,309	0	0	1,309	0
Gray–Scott	1,309	0	0	1,309	0
Coupled map	1,309	0	1,309	0	0
Activator–inhibitor	1,309	0	0	1,309	0
FitzHugh–Nagumo	1,309	0	0	1,309	0
Kuramoto–Sakaguchi	1,309	0	0	1,309	2
Damped symplectic	1,309	0	0	1,309	0
Conservative graph flux	1,309	1,309	0	0	0
Total	13,090	1,309	1,309	10,472	2

The two confirmed strange attractors are a subset of the Kuramoto–Sakaguchi persistent or complex screen. Both candidates selected for definitive testing passed the full confirmation protocol. The coupled-map population comprises 1,309 oscillator-screen candidates; definitive long-horizon periodicity testing remains required before describing them as true oscillators. The full machine-readable table is available in the [dynamical population summary](#).

Conclusions

We represented drug molecules as connected graph cellular automata in which atoms acted as cells, covalent bonds defined local neighbourhoods, and 16-channel atom states evolved through learned transition rules. Complete trajectories were pooled into molecular fingerprints and linked to CYP1A2, CYP2C9, CYP2D6, and CYP3A4 pIC50 values through differentiable ridge regression. Five transition-rule families and two temporal representations were first combined as DS-GCAE, after which endpoint-specific cross-fitted ridge stacking produced CFT-DS-GCAE. On the sealed scaffold holdout, the latter achieved MA-ST-RAE 0.7739 and RMSE 0.8586 pIC50. Its official OpenADMET blind evaluation returned MA-ST-RAE 1.0120, showing a small improvement over the first submission while also revealing a substantial internal-to-blind generalisation gap.

The retained cellular-automata histories also exposed several forms of emergent molecular information dynamics. Across 13,090 screened trajectories we observed contraction towards a point attractor, period-two oscillator candidates, persistent complex motion, and two Kuramoto–Sakaguchi trajectories that satisfied our operational tests for hyperchaotic strange attractors. Their perturbation sensitivity was continually regenerated after renormalization, their leading Lyapunov spectra contained several positive exponents, and perturbed initial conditions remained bounded while approaching the same attracting distribution. Structural association and intervention experiments further indicated that molecular connectivity and bond identity influence the strength of this instability.

The present predictive results leave considerable scope for improved calibration and generalisation, while the dynamical findings establish a practical framework for studying how local chemical interactions generate global computational behaviour. Future development will refine the predictive ensemble, expand confirmatory testing across more molecules and transition rules, and examine whether particular scaffolds, bond arrangements, or chemical encodings reproducibly select point, periodic, complex, or strange-attractor regimes. This combination of molecular prediction and measurable nonlinear dynamics offers a distinctive way to investigate the flow of chemical information through molecular graphs.